THE KERNEL OF THE CHANGE OF BASE FIELD ON CHOW GROUPS IS TORSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to a finite field extension and the computation of the composite of pullback with pushforward as multiplication by the degree.

Editorial note. The label lemma-kernel-to-closure names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-kernel-to-closure –start 2384 –end 2400 –local-id 28171 returned RESULT=OK. Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Flat pullback along a finite flat morphism followed by pushforward is multiplication by the degree. Chow Homology, the lemma computing the composite of flat pullback and proper pushforward along a finite flat morphism as multiplication by the degree of the morphism. Shows that the composite of the change of field map with its pushforward is multiplication by the degree of the field extension.
- The colimit description of Chow groups under an algebraic extension. Same chapter, the construction of the map on Chow groups along an algebraic field extension. Reduces the statement for an arbitrary algebraic extension to the statement for a finite extension.

Source reference. Original lemma 02RH: sources/stacks-chow/tags.txt, source lines 3624–3624. Prerequisite pointer invoked at proof line(s) 2399; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0FH6: sources/stacks-weil/pinned-original-source.tex, source lines 2313–2313. Prerequisite pointer invoked at proof line(s) 2389; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:weil.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let K/k be an algebraic extension of fields. Let X be a finite type scheme over k. The kernel of the map $\mathrm{CH}_i(X) \to \mathrm{CH}_i(X_K)$ constructed in Lemma 0FH6 is torsion.*

Proof. It clearly suffices to show that the kernel of flat pullback $\mathrm{CH}_i(X) \to \mathrm{CH}_i(X_{k'})$ by $\pi : X_{k'} \to X$ is torsion for any finite extension k'/k. This is clear because $\pi_*\pi^*\alpha = [k' : k]\alpha$ by Chow Homology, Lemma 02RH. □